



Analytics Engineering

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RECEIPTIQ:
PRIVACY-FIRST
FINANCE
AGENT
TECHNICAL
CONTRIBUTION
REPORT

Information Pipeline

- **Data Sanitization:** Automated conversion of unstructured OCR strings into validated numerical formats.
- **Temporal Analysis:** Engineered features to extract weekday trends and spending velocity.
- **Privacy Integrity:** Designed processing layers to function entirely within the local environment.

PROCESS_ENGINE.PY

```
# Cleaning & Type Casting
df['total'] = df['raw'].apply(lambda x:
clean_curr(x))

# Feature Engineering
df['day'] = df['date'].dt.day_name()
df['is_weekend'] = df['day'].isin(['Sat',
'Sun'])
```

Pattern Recognition

- **Unsupervised Clustering:** Implemented K-Means to segment transaction history into distinct spending personas.
- **Anomaly Detection:** Integrated Isolation Forest algorithms to flag statistical outliers in expense data.

```
ANOMALY_DETECTOR.PY
from sklearn.ensemble import
IsolationForest

model = IsolationForest(n_estimators=100)
df['anomaly_score'] =
model.fit_predict(df[['total']])

# Highlight unusual spikes
outliers = df[df['anomaly_score'] == -1]
```

Analytical Results

By converting raw extraction into **intelligence**, my contribution allows the user to:

- Identify **leaks** in recurring subscription categories.
- Receive **proactive alerts** on abnormal transaction sizes.
- Forecast **liquidity** for the upcoming month based on historical burn rates.

"The analytics layer bridges the gap between seeing a receipt and understanding financial health."